MILITARY SPECIFICATION

ALTIMETER, PRESSURE AAU-27/A

This amendment forms a part of MIL-A-83212A(USAF), dated 28 October 1976 and is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1: Under "SPECIFICATIONS, Military", the title of "MIL-L-25467" should be changed to read "Lighting, Integral, Red, Aircraft Instrument, General Specification for".

Boxed paragraph at bottom of page: Delete "ASD/ENESS, Wright-Patterson AFB OH 45433" and substitute "Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145".

PAGE 2

Under "STANDARDS, Military" change "MIL-STD-100" to read "DOD-STD-100".

PAGE 3

3.1: Delete paragraph and substitute the following:

"3.1 First article. When specified (see 6.5), the contractor shall furnish sample part(s) for first article testing and approval (see 4.4)."

PAGE 4

3.4.2.2: Delete paragraph and substitute the following:

"3.4.2.2 Mean time between failures (MTBF). The altimeter shall demonstrate a lower test MTBF (θ_1) of 1000 hours to a 90 percent confidence level."

PAGE 9

3.6, line 3: Delete "MIL-STD-100" and substitute "DOD-STD-100".

PAGE 12

3.8.4.1, middle column of table: Add "Height" just above "Inch ± 0.015".

PAGE 14

4.2,a: Delete "Preproduction" and substitute "First article".

PAGE 15

4.4 through 4.5: Delete "Preproduction" wherever mentioned, and substitute "First article".

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PAGE 27

4.6.22.1: Delete paragraph and substitute the following:

"4.6.22.1 First article. The first article reliability test shall be conducted in accordance with MIL-STD-781, using test conditions of paragraph 4.6.22.3 Reliability test conditions, herein which shall demonstrate an altimeter lower test MTBF (θ_1) of 1000 hours to a 90 percent confidence level. The altimeter shall be pneumatically cycled from minimum to maximum altitude and return to minimum (one cycle) within approximately 15 minutes for a total of at least four cycles per hour. During the external vibration period (10 minutes of each hour), the cycling rate shall be increased to 20,000 fpm. Period B shall be approximately 4 hours in duration (see figure 4). During the cooling period, the power shall be turned off and the altitude input returned to sea level. The baro set knob shall be rotated once throughout the complete range each 24-hour period. In case of malfunction, the malfunction shall be recorded, the test interrupted while repairs are made, the test then continued from the point of interruption. At approximately 200-hour intervals, the altimeter shall be subjected to the tests specified in 4.6.2, 4.6.2.1, 4.6.2.2, 4.6.3, 4.6.4. 4.6.4.1, 4.6.6.1, and 4.6.6.1.1. At the conclusion of this test, the altimeter shall meet all individual tests. If recalibration, adjustment, or repair is required to pass any tests, a failure shall be charged against this test. A 25-foot out-of-tolerance condition in scale error will be permitted provided rezeroing by use of the baro set system will correct the condition. This rezeroing shall not result in any out-of-tolerance condition. Scale error tests at all test temperatures shall be performed after rezeroing. Rezeroing will be allowed only once per instrument per test. During all cycling, the vibrator shall be operating and any failure shall be charged as a malfunction against this test. The lighting system shall also be operating during cycling and upon completion of the final cycle, the altimeter shall meet the lighting tests specified in 4.6.7, 4.6.12, and 4.6.19. Failure to meet these tests shall be considered a malfunction. Time accrued during operation of the altimeter in previous tests may be counted as part of the time required by this test. This test may be interrupted for normal weekend shutdown. Accept-reject criteria shall be in accordance with MIL-STD-781, test plan IIIC. If five or more failures occur, the altimeter will have failed the test."

Add new figure 4, which forms page 5 of this amendment.

PAGE 28

4.6.22.2: Delete existing paragraph and substitute the following:

"4.6.22.2 Production. The production reliability test shall be conducted in accordance with 4.6.22.1 except accept-reject criteria shall be in accordance with figure 5, herein."

Add new figure 5, which forms page 6 of this amendment.

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4.6.22.3: Delete paragraph entirely and substitute the following:

"4.6.22.3 Longevity. Two of the six first article reliability test samples shall be subjected to a longevity test in accordance with 4.6.22.3.1 and 4.6.22.3.2. and 4.6.22.3.3.

4.6.22.3.1 General. The equipment shall be tested at the specified test level for a period of time equivalent to the life requirement in this specification. The longevity test shall be 3000 operating hours for each sample. Test time accumulated on the equipment during a reliability test may be applied toward the longevity test when the same sample units are used.

4.6.22.3.2 Sample size. At least two (2) samples shall be subjected to the longevity test.

4.6.22.3.3 Evaluation of longevity test . No accept-reject criteria for this test is specified. The evaluation shall be limited to the analysis of all failures, identification of design and pattern failures, and the submission of recommended design changes to the procuring activity."

Add new paragraph as follows:

"4.6.22.5 Reliability test conditons. Reliability test conditions are as follows:

Temperature	-54°C to +55°C (-65°F to 131°F)
Temperature Cycling	Temperature cycling shall be time to stabilize at the high temperature, plus 2 hours.
Vibration	2.2G ± 10% peak acceleration value at any nonresonant frequency between 20 and 60 Hz measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of operating time.
Equipment on-off cycling ...	Equipment off during cooling cycle and on during heating cycle. (see figure 4)
Input voltage	Nominal specified voltage +5 -2%."

4.7: Delete paragraph and substitute the following:

"4.7 Inspection and test. Tests of methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification."

5.1 through 5.1.1: Delete entirely and substitute the following:

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"5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specificed (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Altimeters shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Altimeters shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation application. Not applicable.

5.1.1.4 Packaging (reference definition in MIL-STD-794). Water-vaporproof caps or plugs shall be applied to electrical connectors and other openings of the altimeters. Each unit shall be packaged in accordance with MIL-P-116, method IC-1, one each in a unit container in accordance with PPP-B-636, domestic class. Cushioning, thickness, density and configuration shall insure shock transmission does not exceed peak values in g's established for the altimeter when complete packs are subjected to the rough handling drop tests stipulated by MIL-STD-794."

PAGE 30

6.3, line 2: Delete "STANAG 3319 and ASCC Air Standard 10/7" and substitute "STANAGs 3319 and 3626 and ASCC Air Standards 10/7, 10/60, and 10/62."

Add new paragraph 6.5 as follows:

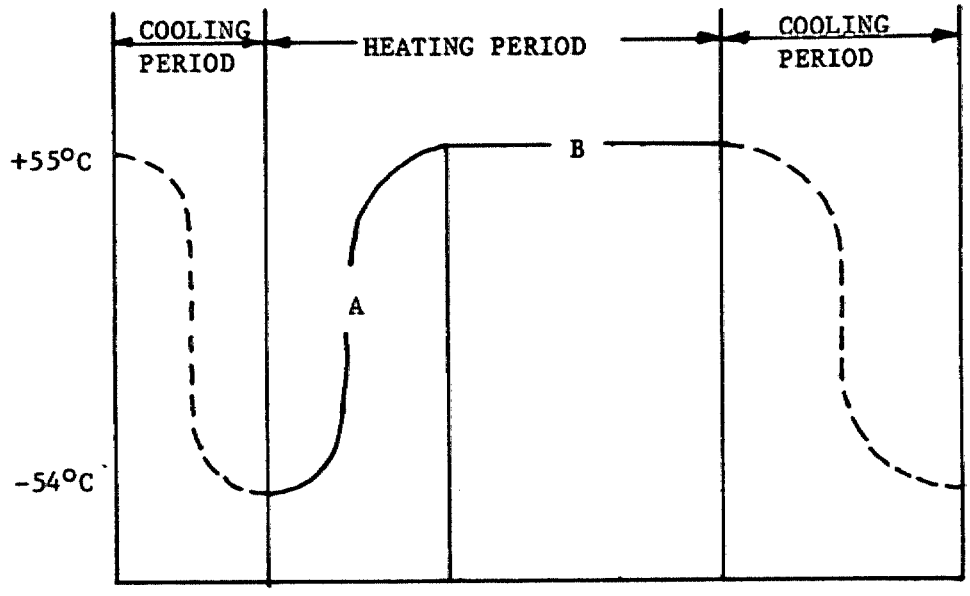
"6.5 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test and approval of the first article."

Custodian
Air Force - 99

Preparing activity:
Air Force - 71

Project number:
6610-F346

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---------- Equipment off

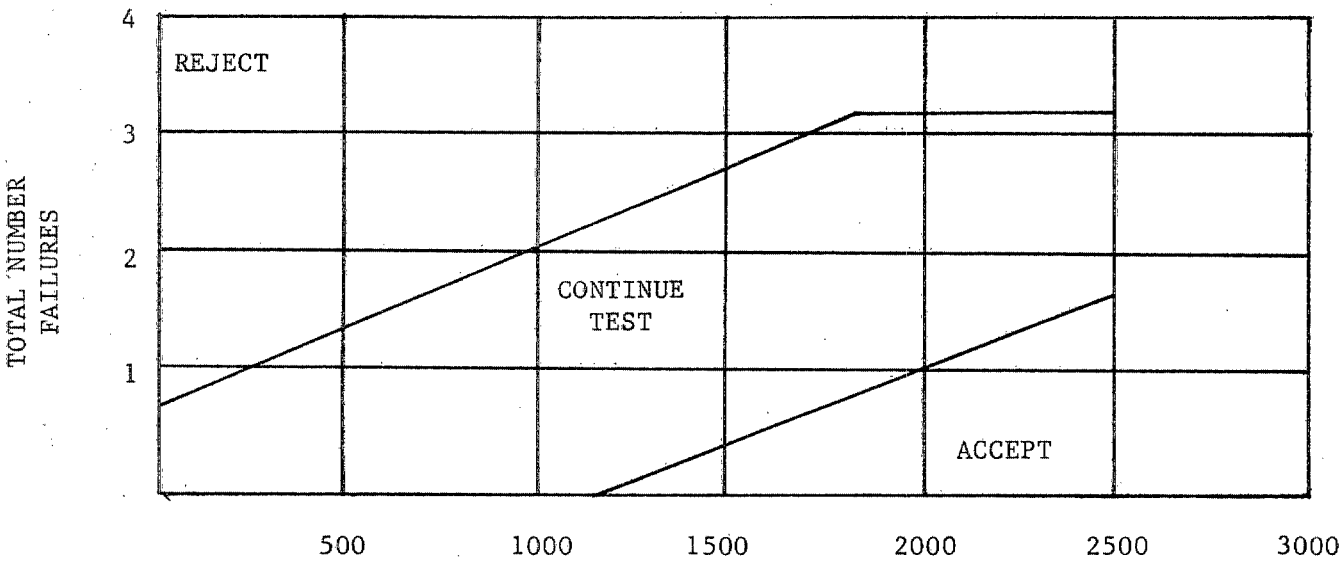
_____ Equipment operated in accordance with duty cycle

A - Time for equipment to stabilize at higher temperature

B - Time of stabilized equipment operation at higher temperature

FIGURE 4. Equipment on-off cycling.

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NO. OF FAILURES	TOTAL TEST TIME (HRS) REJECT (EQUAL OR LESS)	ACCEPT (EQUAL OR MORE)
0	N/A	1100
1	80	1900
2	880	2500
3	1700	2500
4	2500	N/A

FIGURE 5. Accept-reject criteria.

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